Journal of Industrial Ecology / Volume 28, Issue 5 / pp. 1183-1197

RESEARCH ARTICLE | Open Access

Environmental performance and trends of the world's semiconductor foundry industry

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First published: 16 July 2024

https://doi.org/10.1111/jiec.13529

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Abstract

The semiconductor foundry industry faces the challenge of reducing its high environmental impact, mainly due to its energy- and water-intensive processes and significant generation of waste. To date, no other study has focused on the assessment of the environmental performance and related historical trends of this industry as a whole. Methodologically, the first step was to analyze and process a large quantity of economic, production, and environmental data, available in the Corporate Social Responsibility reports of a companies' sample, highly representative of the entire world's foundry industry (about 70% of the global revenue of the related sector). It was thus possible to calculate, using a common manufacturing index (MI) and after appropriate data processing, some key performance indicators, along a significant decade (2012–2021), marked by deep political, economic, and health crises. Some of the main findings of this study are that, over this 10-year period, the increases in technological capacity (patents), wafer production, and revenue (400%, 183%, and 172%, respectively) are matched by a significant increase in hazardous waste generation per MI (20%; 239% in absolute value) and a much larger increase in general

emissions per MI (−32%), mainly due to significant investments in renewable energy sources. The findings of this research could help and guide upcoming sustainability policy decisions and encourage business-to-business collaboration and the adoption of better environmental production practices.

1 INTRODUCTION

Chips are crucial and strategic components for many production sectors, especially, for the hi-tech industry (computers, smartphones, artificial-intelligence [AI] applications, blockchain technology, Internet of things [IoT], and other consumer and industrial equipment and devices) and for technologically supporting new policies for energy transition and environmental and climate mitigation programs (CRS, 2020; McKinsey, 2019; Mordor Intelligence, 2023; SIA, 2021). The semiconductor value chain is extensive, encompassing a range of specialized areas: equipment, electronic design automation (EDA) software, intellectual property (IP) cores, integrated device manufacturers (IDMs) and fabless companies, foundries, and outsourced semiconductor assembly and test (OSAT) services (Deloitte Asia Pacific, 2020; Kleinhans and Baisakova, 2020).

The foundry sector is a fundamental and preponderant part of the semiconductor manufacturing industry that focuses on the production of chips for "fabless" companies (Broadcom, Qualcomm, Nvidia, and MediaTek) that design semiconductors but do not manufacture them (Grimes & Du, 2022). Semiconductor foundry companies ("foundries" or "fabs") have the required and advanced manufacturing technology, know-how, and organizations to produce chips, providing efficient, cost-effective, and high-quality manufacturing solutions for innovative integrated circuits (ICs) (Ciani & Nardo, 2023).

A semiconductor fabless firm, despite not owning manufacturing facilities, could play a crucial role in enhancing sustainability measures such as water recycling, energy use, and waste reduction during the manufacturing process at a semiconductor foundry. Fabless companies, in fact, not having direct control over the manufacturing processes, pursue a path to sustainability working in close partnership with the foundries that manufacture their products. This involvement requires the integration of strategic planning, technological innovation, and collaboration in the whole supply chain, by adopting eco-efficient technologies and processes and investing in R&D to implement eco-design of products characterized by less resource-intensive manufacturing or easier recycling. Industry consortia could help in best practices sharing and to drive sector-wide improvements in sustainability. This could be done through

To improve water recycling, fabless firms could adopt more advanced water treatment technologies, such as ultrafiltration and reverse osmosis, that have been found to significantly increase the recovery rate of high-purity water from the manufacturing process (Tseng et al., 2021). These systems handle wastewater streams (e.g., from wafer back-grinding and sawing processes), reducing the need for freshwater and allowing recovered water to be used in less critical applications (such as electroplating and marking within the foundry) (Baskaran, 2017; Frost & Hua, 2019; Wu et al., 2004). Another study suggests that regression and cost analyses based on mass balance can be applied to optimize the recycling of processing water; this could be extended to the fabless firms, which could coordinate with their foundry partners (Wang et al., 2005).

In terms of energy use, fabless firms can work with foundries to adopt more energy-efficient equipment and processes, possibly supported by data analytics optimizing energy consumption at different stages of production. This could be made using renewable energy sources, investing in energy recovery systems, and implementing smart cluster within the foundry operations (Den et al., 2018). Waste and wastewater generation can be mitigated by material flow analysis to identify areas requiring minimization of resource use and maximization of recycling (Frost & Hua, 2019; Baars et al., 2022). Furthermore, the adoption of eco-design principles in semiconductor sector canreduce waste production (Faraca et al., 2023).

1.1 Background

This industry has developed rapidly over recent years, due to the demand for advanced ICs in many highly profitable and strategic areas, enabling the rapid and comprehensive advancement of knowledge and the improvement of social welfare conditions (Khan et al., 2021; Platzer et al., 2020; Yeung, 2023).

On the other hand, the semiconductor foundry sector has significant environmental impacts; in fact, it requires a significant amount of energy, water (which can put a strain on local water resources, particularly in areas where water is scarce), and various hazardous chemicals (solvents, acids, and gases, such as sulfur hexafluoride, SF6, a powerful greenhouse factor), which could have a significant adverse impact on the environment and risk of exposure for humans (Shen et al., 2018). The related production process also generates greenhouse gas (GHG) emissions and particulate matter (PM), chemical sludge and process water, and general and hazardous waste. To mitigate these environmental effects, fabs have implemented prevention and control countermeasures: energy efficiency solutions; reduction strategies of

Additionally, industry organizations, such as the Semiconductor Industry Association (SIA, 2023), and government agencies, for example, the US Environmental Protection Agency (EPA, 2023a), have approved guidelines and standards to help foundries in their environmental reporting activities and to promote environmental sustainability techniques.

1.2 Scope and aim

By the calculation and analysis of some key performance indicators (KPIs) ofthe Corporate Social Responsibility (CSR) reports data of the examined companies, it is made clear that semiconductor foundries have relevant environmental impact and generate a significant amount of organic and inorganic pollutants and waste, including hazardous waste (heavy metals, alkalis, and acids). These effects result from manufacturing processes, waste disposals, and from the use of water, energy, and various chemicals (Doble & Kumar, 2005). For a complete assessment, from an LCA (life cycle assessment) viewpoint, the effects of the use of electronic devices (e.g., electricity consumption and related GHG emission) should also be contemplated; in addition, the impact from e-waste management, which could be very difficult to recycle and hazardous to the environment if not properly disposed, would also be assessed (Julander et al., 2014; Parajuly et al., 2019).

CSR reports are a precious source of data for stakeholders to assess the performance of a company in relation to sustainability and the environment. In addition, despite certain limitations in readability and transparency, they are for some sectors, as in the case of semiconductor foundry industry, the only available source of data (Wang et al., 2018). It would be desirable for CSR reports, for better accuracy and reliability, to be certified by independent third-party bodies and/or to be arranged according to international standards, such as those issued by the Global Reporting Initiative (GRI) or the Hong Kong Stock Exchange, Ltd. (HKEx) environmental, social, and governance (ESG) rules (Caritte et al., 2015).

By reviewing scientific literature, it is possible to state that, until now, no other research has been conducted regarding the environmental trends assessment of the semiconductor manufacturing or foundry industry as a whole. This could be due to several factors: (1) complexity and multiplicity of wafer production processes that make it impossible to implement a LCA method to this industry; (2) it is a highly strategic industry, considering the strong economic, geopolitical, and military importance of its products, characterized by many technological secrets; (3) it is very difficult to get complete and reliable data for a comprehensive LCA study, even for a single fab; (4) there are no industry environmental performance standards and/or a codified list of universally accepted indicators useful to

manufacturing (Ahmad, 2007; Boyd et al., 2010; Higgs et al., 2009; Huang et al., 2016; Kuo et al., 2022; Liu et al., 2010; Wang, 2014) or are focused only on the carbon footprint of the information and communication technology's (ICT) sector, as has been well clarified by Freitag et al. (2021).

To fill this significant gap, this study aims to investigate the production, technological capacity, and economic and environmental performance trends of several KPIs (revenue, patents, GHG emissions, energy consumption, water withdrawn and wastewater, and general and hazardous waste generation) of the world's major foundries by revenue, during a 10-year period (2012–2021): Taiwan Semiconductor Manufacturing Company Ltd. (TSMC); United Microelectronics Corporation (UMC); GlobalFoundries Inc. (GF); and Semiconductor Manufacturing International Corporation (SMIC).

2 METHODS

This research was carried out according to the following steps: (1) data collection of several environmental items, from CSR reports; (2) calculation of a common manufacturing index (MI) using proper conversion factors (see Table 1); (3) homogenization and arrangement of the various foundries' performance data for the entire decade (see Table 2); and (4) KPIs calculation using a common MI (see Table 4); (5) graphical illustration of KPIs trends and respective discussion.

TABLE 1. World's major semiconductor foundries by revenue (2021).

Company	Headquarters (*a*)	Fabs in operation (country codes) (*b*)	2021 World foundry revenue share (%) (*c*)	Manufacturing indexes (MIs) (*d*)	MI Conversion factor to wfr m^2(*e*)
TSMC	Hsinchu (TW)	TW (10 fabs), CN (2 fabs), US (1 fab), and SG (1 fab)	53	8/12-inch equivalent wafer mask layer	0.0010472 and 0.0023562 *a*
Samsung Foundry	Suwon (KOR)	KOR (4 fabs), CN (1 fab), and US (1	18	8-inch equivalent wafer mask layer	0.0010472

Company	(*a*	(country codes) (*b*)	foundry revenue share (%) (*c*)	indexes (MIs) (*d*)	factor ° to wfr m^2(*e*)
		fab)			
UMC	Hsinchu (TW)	TW (8 fabs), CN (2 fabs), JP (1 fab),	7	wafer-m^2 (wfr-m^2)	1

° From *(d)* to a common MI (wfr-m^2).

a (0.031416 m^2) / 30 (mask layers) = 0.0010472 (wfr-m^2) and (0.070686 m^2) / (30 mask-layers) = 0.0023562 (wfr-m^2).

b (0.031416 m^2) / 30 (mask layers) = 0.0010472 (wfr-m^2).

c (1 m^2) • (10^{-6} mm^2) = 0.0000001 (wfr-m^2).

d (1 m^2) • (10^{-3} cm^2) = 0.0001 (wfr-m^2).

e (31,416 mm^2) • (10^{-6} m^2) = 0.031416 (wfr-m^2).

Sources: Elaboration of data of TrendForce (2023) and of CSR reports (several years) of TSMC, UMC, GF, SMIC, PSMC, and HHS.

TABLE 2. Performance data of the world's major foundries.

	2012	2013	2014	2015	2016	2017	2018
(a) Production (MI)—Wafer-m^2 (in 000) *a*	773.6	898.4	1026.2	,124.1	1228.8	1404.3	1530.0
(b) Revenue (US $ billion) *b*	26.34	30.49	36.17	39.28	43.78	46.30	48.65
(c) Representativeness of the sample (%) *c*	87.0 (76.1)	90.8 (85.0)	90.4 (85.2)	92.5 (87.0)	90.7 (83.0)	91.5 (83.9)	90.6 (73.7)
(d) R&D Expenditures (US $ billion) *b*	2.27	2.63	2.98	3.28	3.61	4.28	4.86

(e) Patents per year	2360	3808	4428	5491	6548	6942	6548
(f) Total patents of the sample [d]	14,389	18,197	22,625	28,116	34,664	41,606	48,154
(g) GHG emissions (kt CO_2e)—Scope 1	2659	2903	3376	3417	3431	3560	3554

[a] Total annual capacity of the related manufacturing facilities of TSMC, UMC, GF, and SMIC. Sum of individual companies' data after homogenization by a common MI.

[b] At current prices in each year.

[c] Based on the total revenue of the world top 10 foundries (for 2015 and 2017, on the total sales of the world top eight foundries) without (or with) the share of Samsung Foundry.

[d] Number of pre-2012 patents: 12,029.

[e] Outsourced energy such as electricity, steam, and heat.

[f] From non-renewable sources. Including electricity and natural gas. Natural gas consumption, expressed in km^3, was converted to GWh assuming a heating value of 39.4 MJ/Sm^3.

[g] The amount of wastewater is a function of tap water consumption and the amount of recycled water.

[h] The item includes: production waste (the most abundant), sludge from water treatment, and domestic waste (the smallest).

Sources: Elaboration of data of CSR reports (several years) of TSMC, UMC, GF, and SMIC; TrendForce (several years); IC Insights (several years); Statista (2022); EPO (2023).

The main available sources of production and environmental data were, if not otherwise indicated, the CSR reports of the analyzed foundries (TSMC, UMC, GF, and SMIC) with their respective affiliates or subsidiaries. In fact, the only data available, regarding the production and environmental impact of these foundries, are those that can be found, albeit with preliminary homogenization processes, in the CSR reports of each company.

Regarding the accuracy of the information reported by the different companies in their respective CSR reports, it should be pointed out that all the examined CSR reports have a high degree of reliability: in fact, the reports of SMIC are certified by the Det Norske Veritas (DNV) Foundation; those of UMC are certified by the Société Générale de Surveillance (SGS); the GF's reports are prepared according to GRI standards and SMIC's reports implement the rules of the HKEx ESG Reporting Guide.

patents were searched according to the following criteria: (1) the search scope delimitation was: “Worldwide—a comprehensive collection of patent applications published from over 90+ countries”; (2) the word “semiconductor” was entered in the string of “Keyword(s) in the title or abstract”; (3) in the search string “Publication date” the various years were entered; and (4) foundries’ names were entered in the string “Inventors or applicants.”

It was not possible to calculate the Samsung Electronics’ performance indicators because, for the years prior to 2020, environmental data, for its foundry division alone, was unobtainable from its CSR reports. In fact, Samsung Electronics, in its reports, groups management and environmental data of different business areas indistinctly: information technology and mobile communications (IM), consumer electronics (CE), and device solutions (DS), which includes Samsung's foundry sector. Powerchip Semiconductor Manufacturing Corp. (PSMC), Hua Hong Semiconductor Ltd. (HSS) Group (2019), and other even smaller ones (for a total revenue share, in 2021, of about 10% of the world's revenue of the whole foundry sector) were not considered due to the unavailability of environmental data for all the years and items analysed. In any case, in 2021, the above mentioned four analyzed companies represent, excluding the share of Samsung Electronics’ foundry business, nearly 90% of the global revenue of the entire sector (71% of share, if the 18% share of Samsung's Foundry is included in the overall total).

Based on the CSR reports data and information, the following items were considered for the subsequent KPIs calculation: (1) Production; (2) Revenue; (3) R&D investments; (4) Patents; (5) direct (Scope 1) and indirect (Scope 2) GHG emissions, referred to in tables and figures as GHG_1 and GHG_2, respectively; (6) Energy consumption; (7) Water (consumption and intake); (8) Wastewater; (9) General waste; and (10) Hazardous waste. Fluorinated-GHG (F-GHG) emissions were not taken into account because the related data are not always available.

Not all the companies report the amount of their Scope 3 GHG emissions. However, the results and assessments on Scope 1 and Scope 2 GHG emissions are not limited and affected by the lack of Scope 3 GHG emissions data because, as is well known, they are a category of emissions that result from activities not directly owned or controlled by the reporting organization and, furthermore, their analysis has little relevance for a paper, like this, that deals with assessing the environmental performance of companies whose production activities, and their environmental impact, are under their control.

As foundries adopt different MIs to which to refer theirperformance, as shown in Table 1, column *(d)*, a preliminary homogenization was therefore necessary, calculating and using different conversion factors (CFs), in order to calculate a singular and common MI, corresponding to the total area of wafers produced in a year (wafer-m^2/y or wfr-m^2/y) and

"12-inch equivalent wafer mask layer," "8-inch equivalent wafer mask layer," and so on), required conversion processes were carried out, considering that: (1) on average, in a single wafer there could be about 30 mask layers (Bosch, 2023; Intel, 2011; UniversityWafer, 2023); and (2) an 11.8-in. wafer (generally indicated as 12″) and a 7.9-in. wafer (generally indicated as 8″) have, correspondingly, a radius of 150 mm (area of 70,686 mm^2) and 100 mm (area of 31,416 mm^2).

In Table 1, the main information of each company, about its MI and the respective conversion factor to the adopted MI (wfr-m^2), are summarized.

3 RESULTS AND DISCUSSION

In Table 2, all the available data, from the analyzed fabs' CSR reports, are shown.

In Table 3, the most significant correlations (Pearson's r—correlation coefficient, PCC), between the annual variations of several items, are shown.

TABLE 3. Indices of significant correlation ($r \geq 70$) between the annual changes of the management variables. Underlying data for this table are available in Table 2 of this article.

(a) Production—Revenue	0.92	(m) GHG_2 - GHGt	0.93
(b) Production—Energy consumption	0.91	(n) Energy—Water consumption	0.76
(c) Production—Water consumption	0.83	(o) Energy—Water withdrawal	0.70
(d) Production—Water withdrawal	0.80	(p) Energy—Hazardous waste	0.81
(e) Production—Hazardous waste	0.87	(q) Energy—Waste (total)	0.92
(f) Production—Waste (total)	0.92	(r) Water consumption—Water withdrawal	0.93
(g) Revenue—Energy consumption	0.79	(s) Water consumption—Hazardous waste	0.90
(h) Revenue—Water consumption	0.72	(t) Water consumption—Waste (total)	0.75
(i) Production—Water withdrawal	0.70	(u) Wastewater—Water withdrawal	0.84
(j) Revenue—Hazardous waste	0.93	(v) Water withdrawal—Hazardous waste	0.78
(k) Revenue—Waste (total)	0.88	(w) Water withdrawal—Waste (total)	0.75
(l) GHG_1 - GHGt	0.71	(x) Hazardous waste—Waste (total)	0.86

TABLE 4. Foundries' key-performance indicators (KPIs). Underlying data for this table are available in Table 2 of this article.

	2012	2013	2014	2015	2016	2017	2018	2019	2
Revenue/Ml (USD/wfr-m^2)	34,050	33,941	35,245	34,943	35,632	32,971	31,799	31,644	3
R&D expenditures on revenue (%)	5.01	5.19	5.20	5.54	5.35	5.77	5.83	6.13	5
R&D investments/Ml (USD/wfr-m^2)	2933	2930	2907	2916	2936	3047	3175	3021	2
GHG_1/Ml (kg CO_2e/wfr-m^2)	3438	3231	3290	3039	2792	2535	2323	2210	1
GHG_2/Ml (kg CO_2e/wfr-m^2)	6709	6343	6367	6167	6352	6436	6090	6154	5
$GHG_{1\&2}$/Ml (kg CO_2e/wfr-m^2)	10,147	9574	9657	9207	9145	8972	8413	8364	7
GHG_1/revenue (kg CO_2e/1000 US $)	101.0	95.2	93.3	87.0	78.3	76.6	73.0	69.8	5
GHG_2/revenue	197.0	186.9	180.6	176.5	177.7	194.4	191.5	194.5	1

By using the graphically represented indicators of Tables 1 and 2, it was possible to outline the trends related to the main aspects about the performance of the various companies as a whole sector. Trend lines were not placed in the graphs when their R^2 had values of little significance (<0.5).

Statistical representativeness of the fabs' sample, based on the global revenue of the world's top 10 foundries, is always very high: consistently around 90% (without Samsung Foundry's share) for each year of the decade analysed (Table 2).

Foundries' water production and their annual revenue have grown from 2012 to 2021 at an almost constant annual rate (Table 2), with the exception of the economic stagnation which occurred in 2019.

After 2016, during which the highest profitability of wafer production was reached (over US $ 35,500/wfr-m^2), there was a definite stabilization, for all subsequent years, around an average value of US $ 32,200/wfr-m^2 (Figure 1a). This price slowdown was mainly due to a market oversupply and to increasing competition (Ross, 2019). This oversupply, at first, led to an average reduction of chips prices, and, subsequently, companies had to search new ways to differentiate their production to preserve their profit margins.

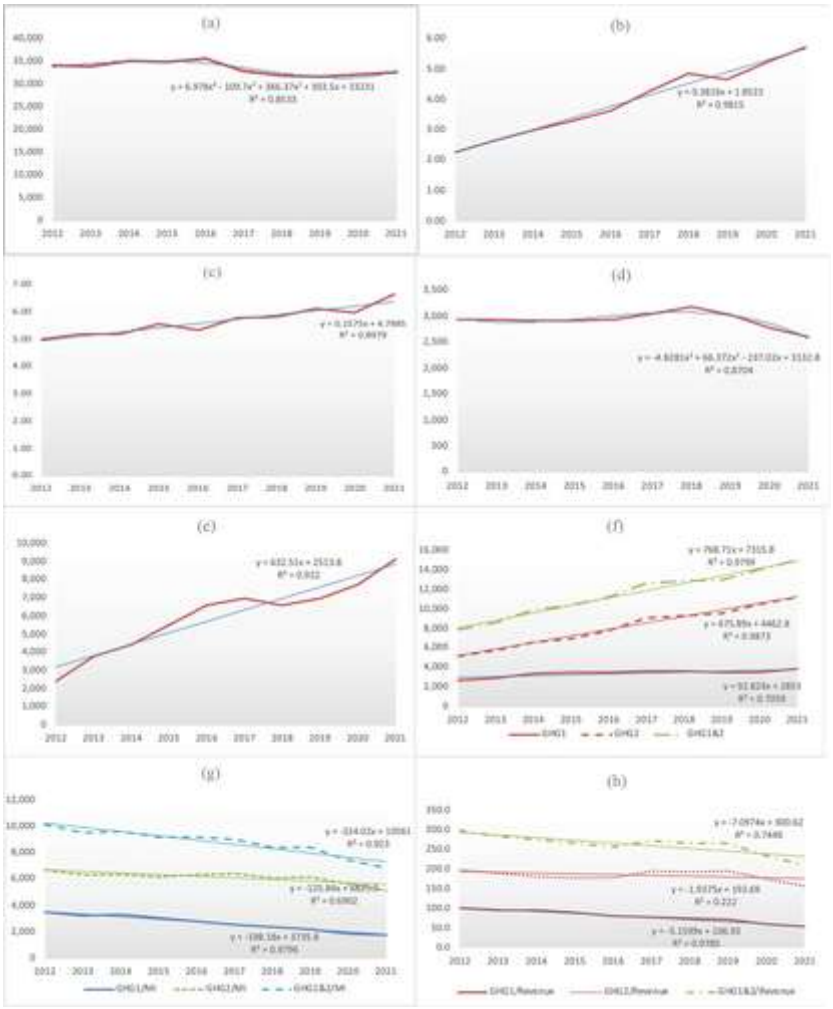


FIGURE 1

Open in figure viewer | PowerPoint

(GHG) emissions (kt CO_2e) (f); GHG emissions/MI (kg CO_2e/wfr-m^2) (g); GHG emissions on revenue (kg CO_2e/1000 US $) (h). Underlying data for this figure are available in Tables 2 and 4 of this article.

About the relationship between environmental management, production, and revenue levels, the environmental performance of a company may significantly affect its revenue. This may be a positive or negative effect depending on the company's environmental management. In fact, a good environmental performance can lead to direct economic benefits: companies that invest in cleaner technologies and energy-efficient processes tend to get reduced operating costs in the long term. Zeng et al. (2010) found that clean production practices may have an overall positive impact on a firm's financial performance. This suggests that investments in environmental sustainability may reduce costs and improve operational efficiency, leading to higher profit margins (Gotschol et al., 2014; Schaltegger & Synnestvedt, 2002).

In addition, also for 2019, R&D investments decreased: from US $4.86 billion to US $4.67 billion (Figure 1b). In fact, the semiconductor market declined in 2019 due to a combination of factors, including: (a) slowdown in the world's economic growth, which led to a reduction of overall demand for electronics and semiconductors (Gopinath, 2019); (b) United States–China and China–Taiwan tensions resulting in increased tariffs on electronics and microchips, which led to a decreased demand for these products (Chen et al., 2021; JEITA, 2019; Pufal et al., 2023); (c) a higher level of excess inventory, due to weakening demand for electronics and semiconductors (SemiMedia, 2023); (d) weak demand for chips from the automotive sector, due to declining sales and production cuts (Brown et al., 2021); (e) weak demand for chips from the smartphone sector, due to market saturation and other economic factors (Mongardini & Radzikowski, 2020); and (f) implications on the consumer market by the COVID-19 stringent political measures (Bauer et al., 2020a). These contractions were anticipated in the previous year (2018) by a significant diminution of new patents, decreasing from 6942 to 6548 (Figure 1b).

R&D investments were gradually increased throughout the decade in relation to revenue (Figure 1c). From the high value of the multiple regression coefficient (R^2) of the trend line, it may be inferred, as it is obvious for an industry with high technological innovation, that these rates will also grow in the next years. R&D expenditures also increased, albeit with some interludes of stagnation in the period 2015−2018, correspondingly with the related production slowdown (Figure 1d).

These investments had strategic implications in the semiconductor foundry industry, characterized by rapid technological change and intense competition. Companies spent a lot on R&D to stay ahead of global competition and to develop new and advanced technologies.

metal–organic vapor deposition (MOVD), nanoimprinting lithography (NIL), 3D nanofabrication, and so on, which have enabled high volume manufacturing (HVM) of smaller and more complex microchips; and (b) environmental sustainability, recycling, and reducing waste, to minimize their environmental impact (Chen et al., 2020; Levinson & Brunner, 2018; Schoot & Schift, 2017; Sharma et al., 2022).

Patents play another important role in the semiconductor industry, generating a competitive advantage and protecting companies' intellectual property. TSMC, known for its advanced manufacturing process, has a strong patent portfolio: about 60% (43,158 patents in 2021) of the total number of patents of all sample companies. This fab, in fact, has a long history of innovation and has invested heavily in R&D to maintain its leading position in this field. The total number of patents has steadily increased over time, with the exception of 2018, when there was a sharp decline from the previous year, due to the above-mentioned general crisis (Figure 1e).

Analyzing the correlations between the annual changes in the multiple variables (Table 3), it was found that the interrelationships between production; revenue; water and energy consumption; and gaseous, liquid, and solid waste generation were very evident. These relationships emerged because each aspect of semiconductor manufacturing was closely related to and often dependent on the other. In fact, manufacturing in the semiconductor industry is notoriously resource intensive, both in terms of energy and water, which are needed to maintain controlled environments such as cleanrooms and for the production of ultra-pure water (UPW), which is essential for washing wafers, removing debris (silica, particles, ions, etc.), and ensuring the highest-quality products (Wishnick, 2021). Den et al. (2018) stated that the semiconductor industry has shown a responsible water usage to maintain business competitiveness and effectively manage water scarcity risks. Energy consumption is correlated not only with production efficiency, but also with the environmental impact of the industry. Zhao and Feng (2021) showed a high correlation between energy consumption and exhaust emission and solid waste generation. In turn, energy use is often influenced by the need to treat and dispose of the waste generated, thus establishing a vicious cycle between energy consumed and waste generated. A company's revenue is directly affected by such dynamics as energy, water expenses, and costs for waste management and compliance with environmental regulations. All of this affects profit margins. Increased operational efficiency, which reduces water and energy consumption and minimizes waste generation, may therefore lead to revenue growth through reduced operating costs and an improved corporate identity linked to its greater environmental responsibility.

3.2 GHG emissions

the current share of ITC sector emissions is between 2.1% and 3.9% of global GHG emissions.

Scope 1 and Scope 2 GHG emissions are counted within specific organizational boundaries as follows: (1) Scope 1 emissions, or direct emissions, are defined as emissions from sources owned or controlled by the organization. They include emissions from combustion in sources owned or under the organization's control, such as boilers, furnaces, vehicles, etc., and those from chemical production in process equipments that are owned or controlled by the organization. The boundary for Scope 1 is the physical location or sites controlled by the company, including buildings, manufacturing facilities, and company vehicles. (2) Scope 2 emissions (energy indirect emissions) result from the generation of purchased electricity, heat, steam, or cooling that the organization consumes. While the emissions physically occur at the facility where the energy is generated, they are accounted for in Scope 2 because they are a result of the organization's energy use. The boundary for Scope 2 includes all locations where the organization consumes purchased energy.

Total GHG emissions almost doubled over the 2012–2021 period: from 7850 to over 15,000 kt CO_2e (Table 2 and Figure 1f). This could be attributed, almost exclusively, to the increase of Scope 2 GHG emissions; while the Scope 1 component has remained basically anchored around the average value of 3400 kt CO_2e per year. Since Scope 1 GHG includes, as well known and stated above, emissions from sources controlled by a company (e.g., fossil fuels used) and GHG Scope 2 includes emissions related to energy purchased by a company, primarily for electrical consumption (by fuels burned by third parties), it means that, during the analyzed period, fabs have increasingly used electricity generated by others to power their production processes. Comparing these emissions with the production index (Figure 1 g) and revenue (Figure 1h), it can be seen that all three lines in the graph have a corresponding trend line with a negative determinant, which is more pronounced for $GHG_{1\&2}$ emissions data. This has occurred because foundries, over the years, have gradually implemented various strategies (such as using biodegradable materials and renewable energy sources, improving energy efficiency, and reducing the use of chemicals that increase GHG emissions) and invested in carbon capture and storage technologies (Crawford et al., 2021).

One of the leading strategies has been the optimal application of control technologies, which have integrated the efficient use of exhaust gas destruction equipment, particularly for critical processes such as etching and thin-film deposition (Zhu et al., 2023). Another major key factor has been the semiconductor industry's voluntary commitment to reduce emissions of perfluorinated compounds (PFCs) since 2010 through collaboration among manufacturers, both with each other and with fabless firmssharing knowledge on effective techniques and strategies (Illuzzi & Thewissen, 2010). This is an excellent example of what can be accomplished

reduction technologies and strategies (Lee et al., 2023). Lastly, pollution taxes and/or subsidies has stimulated semiconductor factories to reduce their GHG emissions (Chi & Hsu, 2017).

3.3 Energy efficiency and intensity

Semiconductor foundries consume relatively high energy per MI in their manufacturing processes. This huge energy consumption, which nearly tripled from 2012 to 2021 (Figure 2b), was related to the need for high temperatures, high-precision air, and humidity control in large clean rooms. Other reasons were the related large-scale manufacturing processes, considering that leading foundries' scale production is about 100,000 (wafer starts per week (WSpW) (Bauer et al., 2020b; Burkacky et al., 2023a). Energy efficiency could also be expressed through energy intensity, calculated as units of energy divided by revenue. This indicator measures the capacity of a company to "convert" energy into revenue: high energy intensity reveals high inefficiency (and high costs) in "converting" energy into revenue. The decrease of energy use per unit of revenue—except in 2020 and 2021, when the global foundry industry had a slowdown in the rate of increase in energy efficiency (12,662 MWh/wfr-m^2 in 2021) and intensity (387 MWh/US $ million in 2021), after a long period of almost constant growth (from 2013 to 2019) (Figure 2a)—was due to moreefficient manufacturing processes and the use of renewable energy (Favino, 2023). Additionally, the industry has also investigated alternative manufacturing technologies, such as those based on molecular and nanoscale engineering (e.g., directed self-assembly or DSA and block copolymer or BCP technologies), which may lead to the development of more energy-efficient manufacturing processes in the future (Chen & Xiong, 2020; Mullen & Morris, 2021). These efforts may also mitigate the impact of increasing demand for semiconductors both on energy use and GHG emissions. Energy efficiency, related to the energy used per unit of production, remained essentially unchanged around the average value of 13,350 MWh/MI (Figure 2b).

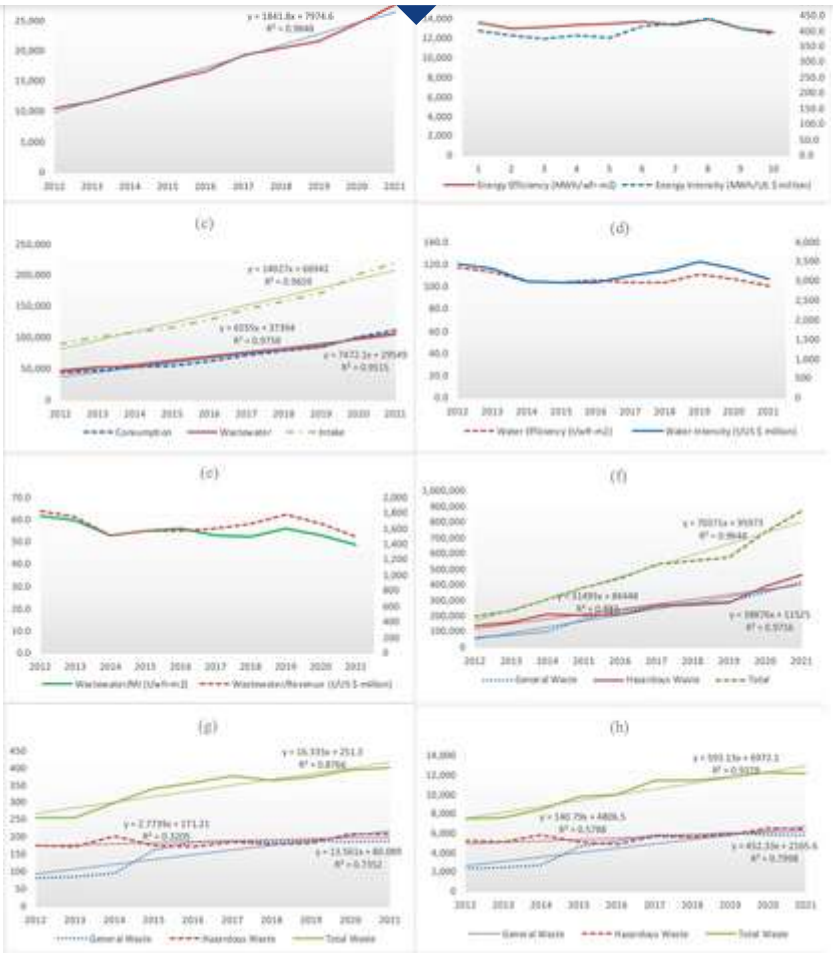


FIGURE 2

Open in figure viewer | PowerPoint

Energy consumption (GWh) (a); energy efficiency and intensity (b); water use (kt) (c); water efficiency and intensity (d); wastewater on manufacturing index (MI) and on revenue (e); waste generated (t) (f); waste generated per MI (kg/wfr-m^2) (g); waste generated per revenue (kg/US $ million) (h). Underlying data for this figure are available in Tables 2 and 4 of this article.

3.4 Water use and wastewater production

Semiconductor foundries require huge amounts of water for their manufacturing processes (Wishnick, 2021). Water consumption in 2021, which has more than doubled since 2012, was nearly 220 kt, almost equal to the average daily water consumption of 28.9 million US citizens (EPA, 2023b) (Figure 2c).

ventilation, and other purposes (Multani, 2020). The amount of water withdrawal for these purposes changes significantly depending on the complexity of a process. Some foundries, like TSMC, use recirculated water systems in combination with alternative water sources (such as rainwater or wastewater) to minimize their water withdrawal, but other fabs still use large amounts of freshwater with a significant impact on local water resources (Den et al., 2018; TSMC, 2023).

Over the years, the amount of water intakes, in relation to MI and revenue, has remained basically stable around the average value of 107 t (metric tons)/wfr-m^2 and 3200 t/US $ million (Figure 2d). Since 2017, the two corresponding trend lines diverge: that of the water withdrawal/MI ratio is clearly lower than that of the water/revenue ratio. This is due, fundamentally, to the lower unit profitability of wafers produced related to the performance of the revenue/MI ratio.

Semiconductor foundries generate a significant amount of wastewater during manufacturing processes, which may contain various chemicals and hazardous substances. The treatment and disposal of this wastewater is fundamental to safeguard the environment and public health. Typically, in the foundries, wastewater treatment involves physical, chemical, and biological methods to remove contaminants and pollutants before discharge (Wang et al., 2023). Some foundries also implement closed-loop systems, such as zero liquid discharge (ZLD) systems, to minimize wastewater discharge and reduce its environmental impact (Burkacky et al., 2023b; Gandhi, 2020; Makini, 2015).

Wastewater generation has slightly decreased. In fact, the relative trend line has a negative determinant (m = −0.9523) (Figure 2e) because some companies, such as TSMC, have implemented systematic and resolute programs for water reuse and recycling. TSMC, in fact, for every year of the period, has a wastewater/MI index that is constantly about 60% of the corresponding average of the all examined companies.

3.5 Waste generation

Foundries generate chemical, electronic, and hazardous substances. Hazardous waste generation is due to the use of toxic chemicals in the manufacturing process: additives, photoactive compounds, polymer, salts, etc., which are mainly used in photolithography, in the liquid or gaseous phase. They, sometimes secretly, may include solvents, acids, and heavy metals that could be harmful to the environment and human health if not properly managed (Kim et al., 2018).

in part to the increased manufacturing complexity of the new wafers (with smaller technology nodes), which required much more water per unit. The most relevant aspect, however, is not only the fact that hazardous waste has more than doubled in absolute values since 2012, but that the corresponding trend line has shown a conspicuous increase since 2019.

General and hazardous waste generation per MI has also strongly increased, over the years, in relation to wafers produced and annual revenue (Figure 2g,h). Since 2017, however, there is a settling trend in the rise of the trend lines of waste generated per MI and waste generated per revenue, due to the commissioning of new and modern plants and/or to the startup of more technologically advanced production lines.

Added to this, there is the consideration that, in the semiconductor manufacturing sector, the environmental impact of waste may be more severe, mainly due to the increased use of materials with uncertain long-term environmental effects. In particular, the relationship between the amount of these materials and the related environmental impact is not so evident. For instance, the impact could be exponential and not linear to the increase in the amount of waste. The chemical waste management in the semiconductor industry is very complexfor the various toxic components involved. This highlights the need of addressing these environmental challenges through comprehensive strategies for reducing the use of hazardous materials, improving waste treatment technologies, and implementing strict environmental safety measures and protocols (Shen et al., 2018).

To reduce waste generation, some semiconductor foundries have implemented sustainability practices: a strategy to reduce the use of hazardous chemicals; recycling waste products; waste minimization processes, like the reduced use of test wafers; adopting new techniques (such as more advanced NIL and EUVL), and improving energy efficiency (CRS, 2020; Mullen & Morris, 2021). For the future, this sector will have to search and develop alternative production processes to reduce waste generation and promote sustainability (Englund et al., 2023; Shen et al., 2018).

4 CONCLUSIONS

In absolute value, all the main indicators of the examined sample of foundry companies have shown decidedly growing trends for the entire period. In fact, since 2012, increases in wafer production (183%), revenue (172%), and technological capacity (400% more patents) have been matched by corresponding increases in resource use and environmental impacts, in terms of energy consumption (164%), water use (163%), and by more than proportional increases in general waste (568%) and hazardous waste (239%) generation. These negative findings have

renewable energy sources, such as photovoltaic and wind power, which are very helpful in improving energy efficiency and reducing carbon footprint.

The increase in waste generation is the most notable of these facts, and it is even more remarkable considering that general waste and hazardous waste are, among all these, the only two factors whose trends have significantly grown also in relation to fabs' MI and revenue. It is, therefore, critical for this industry to reduce its environmental impact and promote more sustainable practices to mitigate the negative effects of microchip production.

The findings of this research could help and guide upcoming sustainability policy decisions and stimulate the adoption of better environmental production practices, for instance, by encouraging the development and implementation of waste reduction technologies, improving better wastewater management, providing incentives for better efficient production practices, promoting renewable energy investments, and establishing new benchmarks or promoting technological innovations.

FUNDING INFORMATION

No funds to declare.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

Ahmad, A. H. (2007). *Life cycle assessment in semiconductor foundry*. University of Southern Queensland. https://eprints.usq.edu.au/3939/1/Ahmad_2007.pdf

Google Scholar | Find PDF

Google Scholar | Find PDF

Baars, J., Rajaeifar, M. A., & Heidrich, O. (2022). Quo vadis MFA? Integrated material flow analysis to support material efficiency. *Journal of Industrial Ecology*, **26**, 1487–1503. https://doi.org/10.1111/jiec.13288

Web of Science® | Google Scholar | Find PDF

Baskaran, A. (2017). Water use in the semiconductor industry. *Sustainalytics.*. https://www.sustainalytics.com/esg-research/resource/investors-esg-blog/waste-not-want-not-water-use-in-the-semiconductor-industry#:~:text=Semiconductor%20manufacturing%2C%20however%2C%20requires%20large,ultra%2Dpure%20water%20per%20day

Google Scholar | Find PDF

Bauer, H., Burkacky, O., Kenevan, P., Mahindroo, A., & Patel, M. (2020a). *Coronavirus: Implications for the semiconductor industry*. McKinsey & Co. https://www.mckinsey.com/industries/semiconductors/our-insights/coronavirus-implications-for-the-semiconductor-industry/it-IT

Google Scholar | Find PDF

Bauer, H., Burkacky, O., Lingemann, S., Pototzky, K., Kenevan, P., & Wiseman, B. (2020b). *Semiconductor design and manufacturing: Achieving leading-edge capabilities*. McKinsey & Co. https://www.mckinsey.com/industries/industrials-and-electronics/our-insights/semiconductor-design-and-manufacturing-achieving-leading-edge-capabilities

Google Scholar | Find PDF

Belton, P. (2021). *The computer chip industry has a dirty climate secret*. The Guardian. https://www.theguardian.com/environment/2021/sep/18/semiconductor-silicon-chips-carbon-footprint-climate

Google Scholar | Find PDF

Bosch. (2023). Semiconductor terminology. BOSCH. https://www.bosch-presse.de/pressportal/de/en/begriffe-rund-um-halbleiter-229060.html#:~:text=Roughly%2030%20layers%20are%20stacked,throu

Google Scholar | Find PDF

Boyd, S. B., Horvath, A., & Dornfeld, D. A. (2010). Life-cycle assessment of computational logic produced from 1995 through 2010. *Environmental Resources Letters*, **5**(1), 1–8. https://doi.org/10.1088/1748-9326/5/1/014011

Google Scholar | Find PDF

Brown, D., Flickenschild, M., Mazzi, C., Gasparotti, A., Panagiotidou, Z., Dingemanse, J., & Bratzel, S. et al. (2021). *The future of the EU automotive sector*. European Parliament's Committee on Industry, Research and Energy (ITRE). https://www.europarl.europa.eu/RegData/etudes/STUD/2021/695457/IPOL_STU(2021)695457_EN.pdf

Google Scholar | Find PDF

Burkacky, O., Dragon, J., & Lehmann, N. (2023a). *The semiconductor decade: A trillion-dollar industry*. McKinsey & Company. https://www.mckinsey.com/industries/semiconductors/our-insights/the-semiconductor-decade-a-trillion-dollar-industry

Google Scholar | Find PDF

Burkacky, O., Nikolka, M., Göke, S., Patel, M., & Spiller, P. (2023b). *Sustainability in semiconductor operations: Toward Net-Zero production*. McKinsey & Co. https://www.mckinsey.com/industries/semiconductors/our-insights/sustainability-in-semiconductor-operations-toward-net-zero-production

Google Scholar | Find PDF

Caritte, V., Acha, S., & Shah, N. (2015). Enhancing corporate environmental performance through reporting and roadmaps. *Business Strategy and the Environment*, **24**(4), 289–308. https://doi.org/10.1002/BSE.1818

Google Scholar | Find PDF

Chen, J., Lin, H., & Lien, Y. (2021). *Taiwan's shifting role in the global supply chain in the US-China trade war*. Joint U.S.-Korea Academic Studies. https://keia.org/wp-content/uploads/2021/07/KEI_JointUS-Korea_2021_3_14.pdf

Chen, X., Despeisse, M., & Johansson, B. (2020). Environmental sustainability of digitalization in manufacturing: A review. *Sustainability*, **12**(12), 10298. https://doi.org/10.3390/su122410298

Google Scholar | Find PDF

Chen, Y., & Xiong, S. (2020). Directed self-assembly of block copolymers for sub-10 nm fabrication. *International Journal of Extreme Manufacturing*, **2**(2), 032006. https://doi.org/10.1088/2631-7990/aba3ae

CAS | Google Scholar | Find PDF

Chiang, C., & Hsu, H.-L. (2017). Incorporating pollution taxes and/or subsidies into master planning in semiconductor foundry plants. *International Journal of Management and Sustainability*, **6**(1), 8–22. https://doi.org/10.18488/JOURNAL.11.2017.61.8.22

Google Scholar | Find PDF

Choi, K.-M. (2018). Airborne PM2.5 characteristics in semiconductor manufacturing facilities. *AIMS Environmental Science*, **5**(3), 216–228. https://doi.org/10.3934/environsci.2018.3.216

CAS | Google Scholar | Find PDF

Choi, S., Yoon, C., Kim, S., Kim, W., Ha, K., Jeong, J., Kim, J., Shin, J., & Park, D. (2018). Comprehensive evaluation of hazardous chemical exposure control system at a semiconductor manufacturing company in South Korea. *International Journal of Environmental Research and Public Health*, **15**(6), 1162. https://doi.org/10.3390/ijerph15061162

PubMed | Google Scholar | Find PDF

Ciani, A., & Nardo, M. (2023). *The position of the EU in the semiconductor value chain: Evidence on trade, foreign acquisitions, and ownership*. European Commission. Ispra JRC129035. https://joint-research-centre.ec.europa.eu/system/files/2023-04/JRC129035.pdf

Google Scholar | Find PDF

-with-its-giant-carbon-footprint?leadSource=uverify%20wall

Google Scholar | Find PDF

CRS (Congressional Research Service). (2020). *Semiconductors: U.S. industry, global competition, and federal policy*. The Free Library. CRS Reports and Issue Briefs. https://www.thefreelibrary.com/Semiconductors%3a+U.S.+Industry%2C+Global+Competition%2C+and+Federal+Policy.-a0661274213

Google Scholar | Find PDF

Deloitte Asia Pacific Ltd. (2020). Rise of the Big 4. The semiconductor industry in Asia Pacific. https://www2.deloitte.com/content/dam/Deloitte/cn/Documents/technology-media-telecommunications/cn-tmt-rise-of-the-big-4-en-082820.pdf

Google Scholar | Find PDF

Den, W., Chen, C.-H., & Luo, Y.-C. (2018). Revisiting the water-use efficiency performance for microelectronics manufacturing facilities: Using Taiwan's Science Parks as a case study. *Water-Energy Nexus*, **1**(2), 116–133. https://doi.org/10.1016/j.wen.2018.12.002

Google Scholar | Find PDF

Doble, M., & Kumar, A. (2005). Chapter 14 - Semiconductor waste treatment. *Biotreatment of Industrial Effluents*, **2005**, 157–168. https://doi.org/10.1016/B978-075067838-4/50015-4

Google Scholar | Find PDF

Englund, E., Eliadis, E., Flynn, C., & Lubowe, D. (2023). *How semiconductor companies can lead on sustainability issues*. https://www.ey.com/en_us/tmt/how-semiconductor-companies-can-lead-on-sustainability

Google Scholar | Find PDF

EPA (US Environmental Protection Agency). (2023a). *Semiconductor manufacturing*. https://ordspub.epa.gov/ords/guideme_ext/f?p=guideme:gd:semicond

EPA (US Environmental Protection Agency). (2023b). *WaterSense: Statistics and facts*. https://www.epa.gov/watersense/statistics-and-facts

Google Scholar | Find PDF

EPO (Europaen Patent Office). (2023). *Espacenet: Free access to over 140 million patent documents*. https://worldwide.espacenet.com/

Google Scholar | Find PDF

Faraca, G., Spiliotopoulos, C., Ranea Palma, A., Pérez-Camacho, M. N., Alfieri, F., Bernad Beltran, D., Lag Brotons, A., Delre, A., Pérez Arriba, Z., Arcipowska, A., La Placa, M. G., & Wolf, O. (2023). *Ecodesign for sustainable products. Regulation—preliminary study on new product priorities*. *JRC Science for Policy Report. Technical Report (draft) of the European Commission*. https://susproc.jrc.ec.europa.eu/product-bureau/sites/default/files/202301/Preliminary%20ESPR%20WP%20Report_MERGED_CLEAN_.pdf

Google Scholar | Find PDF

Favino, C. (2023). *The role of semiconductors in the renewable energy transition*. Earth Org. https://earth.org/semiconductors/

Google Scholar | Find PDF

Freitag, C., Berners-Lee, M., Widdicks, K., Knowles, B., Blair, G. S., & Friday, A. (2021). The real climate and transformative impact of ICT: A critique of estimates, trends, and regulations. *Patterns*, **2**(9), 100340. https://doi.org/10.1016/j.patter.2021.100340

Web of Science® | Google Scholar | Find PDF

Frost, K., & Hua, I. (2019). Quantifying spatiotemporal impacts of the interaction of water scarcity and water use by the global semiconductor manufacturing industry. *Water Resources and Industry*, **22**, 100115. https://doi.org/10.1016/j.wri.2019.100115

Web of Science® | Google Scholar | Find PDF

Google Scholar | Find PDF

GlobalFoundries. several years. Corporate Social Responsibility (CSR) Reports. https://gf.com/about-us/corporate-responsibility/resources/

Google Scholar | Find PDF

Gopinath, G. (2019). *The world economy: Synchronized slowdown, precarious outlook*. https://www.imf.org/en/Blogs/Articles/2019/10/15/the-world-economy-synchronized-slowdown-precarious-outlook

Google Scholar | Find PDF

Gotschol, A., De Giovanni, P., & Esposito Vinzi, V. (2014). Is environmental management an economically sustainable business? *Journal of Environmental Management*, **144**, 73–82. https://doi.org/10.1016/j.jenvman.2014.05.001

PubMed | Web of Science® | Google Scholar | Find PDF

Grimes, S., & Du, D. (2022). China's emerging role in the global semiconductor value chain. *Telecommunication Policy*, **46**, 101959. https://doi.org/10.1016/j.telpol.2020.101959

Google Scholar | Find PDF

Higgs, T., Cullen, M., Yao, M., & Stewart, S. (2009). Developing an overall CO_2 footprint for semiconductor products. IEEE International Symposium on Sustainable Systems and Technology. Tempe (AZ, USA), 18-20 May 2009. https://doi.org/10.1109/ISSST.2009.5156786

Google Scholar | Find PDF

Hua Hong Semiconductor (HHS) Ltd. (2019). *Annual report*. https://www1.hkexnews.hk/listedco/listconews/sehk/2020/0408/2020040800125.pdf

Google Scholar | Find PDF

284. https://doi.org/10.1007/s13762-015-0869-z

CAS | Google Scholar | Find PDF

Hui, V. W. (2019). *A systematic approach for waste management in the semiconductor industry* [Doctoral dissertation, Faculty of Mississippi State University]. https://scholarsjunction.msstate.edu/cgi/viewcontent.cgi?article=1321&context=td

Google Scholar | Find PDF

IC Insights. several years. *News*. https://www.icinsights.com/news/

Google Scholar | Find PDF

Illuzzi, F., & Thewissen, H. (2010). Perfluorocompounds emission reduction by the semiconductor industry. *Journal of Integrative Environmental Sciences*, **7**(1), 201–210. https://doi.org/10.1080/19438151003621417

Google Scholar | Find PDF

Intel. (2011). *From sand to silicon: Making of a chip*. https://download.intel.com/newsroom/kits/chipmaking/pdfs/Sand-to-Silicon_32nm-Version.pdf

Google Scholar | Find PDF

JEITA (Japan Electronics and Information Technology Industries Association). (2019). *Production forecasts for the global electronics and information technology industries*. https://www.jeita.or.jp/english/topics/2019/1218.pdf

Google Scholar | Find PDF

Julander, A., Lundgren, L., Skare, L., Grandér, M., Palm, B., Vahter, M., & Lidén, C. (2014). Formal recycling of e-waste leads to increased exposure to toxic metals: An occupational exposure study from Sweden. *Environment International*, **73**, 243–251. https://doi.org/10.1016/j.envint.2014.07.006

Khan, S. M., Mann, A., & Peterson, D. (2021). *The semiconductor supply chain: Assessing national competitiveness*. CSET (Center for Security and Emerging Technology). https://cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief.pdf

Google Scholar | Find PDF

Kim, S., Yoon, C., Ham, S., Park, J., Kwon, O., Park, D., Choi, S., Kim, S., Ha, K., & Kim, W. (2018). Chemical use in the semiconductor manufacturing industry. *International Journal of Occupational and Environmental Health*, **24**(3-4), 109–118. https://doi.org/10.1080/10773525.2018.1519957

PubMed | Web of Science® | Google Scholar | Find PDF

Kleinhans, J. P., & Baisakova, N. (2020). *The global semiconductor value chain*. Stiftung Neue Verantwortung. https://www.stiftung-nv.de/sites/default/files/the_global_semiconductor_value_chain.pdf

Google Scholar | Find PDF

Kuo, T.-C., Kuo, C.-Y., & Chen, L.-W. (2022). Assessing environmental impacts of nanoscale semiconductor manufacturing from the life cycle assessment perspective. *Resources, Conservation and Recycling*, **182**, 1–9. https://doi.org/10.1016/j.resconrec.2022.106289

Google Scholar | Find PDF

Lee, B.-J., Yun, S.-Y., Jeong, I.-K., Hwang, Y., Park, J.-H., & Kim, J. (2023). Improving the measurement of characteristic parameters for the determination of GHG emissions in the semiconductor and display industries in Korea. *Applied Sciences*, **13**(15), 8834. https://doi.org/10.3390/app13158834

CAS | Google Scholar | Find PDF

Levinson, H. J., & Brunner, T. A. (2018). Current challenges and opportunities for EUV lithography. Proceedings of SPIE 10809. International Conference on Extreme Ultraviolet Lithography. Monterey (CA, USA) 24 October 2018. 2018:1080903. https://doi.org/10.1117/12.2502791

Google Scholar | Find PDF

industry. *Journal of Cleaner Production*, 13(3), 119 https://doi.org/10.1016/j.jclepro.2003.10.004

CAS | Web of Science® | Google Scholar | Find PDF

Makini, B. (2015). *Achieving zero liquid discharge in the semiconductor industry*. Columbia University. https://www.yumpu.com/en/document/download/43244406/7360c-7ff2f-a3ae9-34d5c-5d7d4-20e1b-7e2c5-f2c0f

Google Scholar | Find PDF

McKinsey & Co. (2019). *McKinsey on semiconductors. Creating value, pursuing innovation, and optimizing operation*. https://www.mckinsey.com/~/media/McKinsey/Industries/Semiconductors/Our%20Insights/McKinsey%20on%20Semiconductors%20Issue%207/McK_Semiconductors_Oct2019-Full%20Book-V12-RGB.ashx

Google Scholar | Find PDF

Mongardini, J., & Radzikowski, A. (2020). *Global smartphones sales may have peaked*. International Monetary Fund Working Paper. https://www.elibrary.imf.org/view/journals/001/2020/070/article-A001-en.xml

Google Scholar | Find PDF

Mordor Intelligence. (2023). *Semiconductor foundry market: Growth, trends, Covid-19 impact, and forecasts (2023 - 2028)*. https://www.mordorintelligence.com/industry-reports/semiconductor-foundry-market3

Google Scholar | Find PDF

Mullen, E., & Morris, M. A. (2021). Green nanofabrication opportunities in the semiconductor industry: A life cycle perspective. *Nanomaterials*, **11**(5), 1085. https://doi.org/10.3390/nano11051085

CAS | PubMed | Web of Science® | Google Scholar | Find PDF

Multani, K. (2020). *Water scarcity and the semiconductor industry*. Stanford University. http://large.stanford.edu/courses/2020/ph240/multani2/

Parajuly, K., Kuehr, R., Awasthi, A. K., Fitzpatrick, C., Lepawsky, J., Smith, E., Widmer, R., & Zeng, X. (2019). *Future e-waste scenarios*. StEP (Bonn), UNU ViE-SCYCLE (Bonn) & UNEP IETC (Osaka). https://wedocs.unep.org/bitstream/handle/20.500.11822/30809/FutEWSc.pdf?sequence=1&isAllowed=y

Google Scholar | Find PDF

Platzer, M. D., Sutter, K. M., & Sargent, J. F. (2020). *Semiconductors: U.S. industry, global competition, and federal policy*. US Congressional Research Service. https://sgp.fas.org/crs/misc/R46581.pdf

Google Scholar | Find PDF

Pufal, J., Shun, H., Ben Ammar, R., Jackstadt, J., & Schultheiß, J. (2023). *The Great Shutdown. How disruptive would a military escalation of the China Taiwan conflict be to global economies?*. Technische Hochschule Brandenburg. https://doi.org/10.25933/opus4-2894

Google Scholar | Find PDF

Ross, A. (2019). *Global semiconductor revenue growth in 2018 was weaker than 2017*. Information Age. https://www.information-age.com/global-semiconductor-revenue-growth-13486/

Google Scholar | Find PDF

Ruberti, M. (2022). The chip manufacturing industry: Environmental impacts and eco-efficiency analysis. *Science of The Total Environment*, **858**, 159873. https://doi.org/10.1016/j.scitotenv.2022.159873

PubMed | Google Scholar | Find PDF

Schaltegger, S., & Synnestvedt, T. (2002). The link between 'green' and economic success: Environmental management as the crucial trigger between environmental and economic performance. *Journal of environmental management*, **65**(4), 339–346. https://doi.org/10.1006/JEMA.2002.0555

PubMed | Web of Science® | Google Scholar | Find PDF

Schoot, J. V., & Schift, H. (2017). Next-generation lithography—An outlook on EUV projection and nanoimprint. *Advanced Optical Technologies*, **6**(3-4), 159–162. https://doi.org/10.1515/aot-2017-0040

SemiMedia. (2023). *Consumer electronics chips are oversupplied*. https://www.semimedia.cc/?p=13796

Google Scholar | Find PDF

Sharma, E., Rathi, R., Misharwal, J., Sinhmar, B., Kumari, S., Dalal, J., & Kumar, A. (2022). Evolution in lithography techniques: Microlithography to nanolithography. *Nanomaterials*, **12**, 2754. https://doi.org/10.3390/nano12162754

CAS | PubMed | Google Scholar | Find PDF

Shen, C.-W., Tran, P. P., & Minh Ly, P. T. (2018). Chemical waste management in the U.S. semiconductor industry. *Sustainability*, **10**(5), 1545. https://doi.org/10.3390/su10051545

Web of Science® | Google Scholar | Find PDF

SIA (Semiconductor Industry Association). (2021). *State of the U.S. semiconductor industry*. https://www.semiconductors.org/wp-content/uploads/2021/09/2021-SIA-State-of-the-Industry-Report.pdf

Google Scholar | Find PDF

SIA (Semiconductor Industry Association). (2023). *Environment, health & safety: Support environmental sustainability*. https://www.semiconductors.org/policies/environment-health-safety

Google Scholar | Find PDF

SMIC. several years. *Corporate social responsibility (CSR) reports*. https://www.smics.com/en/site/responsibility_report_honor3

Google Scholar | Find PDF

Statista. (2022). *GlobalFoundries—Statistics & facts*. https://www.statista.com/topics/9869/globalfoundries/#topicOverview

TrendForce. several years. *Press releases: Semiconductors*. https://www.trendforce.com/presscenter/news/Semiconductors

Google Scholar | Find PDF

Tseng, S.-F., Lo, C.-M., & Hung, C.-H. (2021). Evaluation of the benefit of practically operating reverse osmosis system in the factory: Taking the recycling of KI solution and water of the screen polarizing plate as an example. *Water Reuse*, **11**(3), 329–346. https://doi.org/10.2166/wrd.2021.110

CAS | Google Scholar | Find PDF

TSMC. (2023). *Water management*. https://esg.tsmc.com/en/focus/greenManufacturing/waterResourceManagement.html

Google Scholar | Find PDF

TSMC. several years. *Corporate Social Responsibility (CSR) Reports*. https://esg.tsmc.com/en/resources/documents.html

Google Scholar | Find PDF

UMC. several years. *Corporate Social Responsibility (CSR) reports*. https://www.umc.com/en/Download/corporate_sustainability_reports

Google Scholar | Find PDF

UniversityWafer. (2023). *How do you make silicon wafers into PC chips*. https://www.universitywafer.com/how-to-make-silicon-wafers-into-computer-chips.html

Google Scholar | Find PDF

Villard, A., Lelah, A., & Brissaud, D. (2015). Drawing a chip environmental profile: Environmental indicators for the semiconductor industry. *Journal of Cleaner Production*, **86**, 98–109. https://doi.org/10.1016/J.JCLEPRO.2014.08.061

Wang, B., Lee, L., Chen, C., Chang, C., & Chao, A. (2005). Optimization of semiconductor processing water management strategy. *Journal of Environmental Informatics*, **5**, 81–88. https://doi.org/10.3808/JEI.200500049

Google Scholar | Find PDF

Wang, H. C. (2014). *Carbon footprints of different processes in the integrated circuit foundry industry*. National Chiao Tung University. https://ir.nctu.edu.tw/handle/11536/74880

Google Scholar | Find PDF

Wang, Q., Huang, N., Cai, H., Chen, X., & Wu, Y. (2023). Water strategies and practices for sustainable development in the semiconductor industry. *Water Cycle*, **4**, 12–16. https://doi.org/10.1016/j.watcyc.2022.12.001

Web of Science® | Google Scholar | Find PDF

Wang, Z., Hsieh, T.-S., & Sarkis, J. (2018). CSR performance and the readability of CSR reports: Too good to be true? *Corporate Social Responsibility and Environmental Management*, **25**(1), 66–79. https://doi.org/10.1002/CSR.1440

Web of Science® | Google Scholar | Find PDF

Wishnick, E. (2021). Semiconductors and water scarcity in China. *The Diplomat*. https://thediplomat.com/2021/08/water-with-your-chips-semiconductors-and-water-scarcity-in-china/

Google Scholar | Find PDF

Wu, M., Sun, D., & Tay, J. H. (2004). Process-to-process recycling of high-purity water from semiconductor wafer back-grinding wastes. *Resources Conservation and Recycling*, **41**(2), 119–132. https://doi.org/10.1016/J.RESCONREC.2003.09.003

Google Scholar | Find PDF

1080/00130095.2021.2019010?needAccess=true&role=button

Google Scholar | Find PDF

Zeng, S. X., Meng, X. H., Yin, H. T., Tam, C. M., & Sun, L. (2010). Impact of cleaner production on business performance. *Journal of Cleaner Production*, **18**(10-11), 975–983. https://doi.org/10.1016/J.JCLEPRO.2010.02.019

Web of Science® | Google Scholar | Find PDF

Zhao, S., & Feng, H. (2021). Investigation of the 60-year relationship between energy consumption and environmental quality in China. *Environmental Science and Pollution Research*, **29**, 14453–14468. https://doi.org/10.1007/s11356-021-16498-3

PubMed | Google Scholar | Find PDF

Zhu, S., Hu, H., Yang, H., Qu, Y., & Li, Y. (2023). Mini-review of best practices for greenhouse gas reduction in Singapore's semiconductor industry. *Processes*, **11**(17), 2120. https://doi.org/10.3390/pr11072120

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